

Nuclear Toeplitz operators between Fock spaces: an exact sparse-atomic classification

Candidate substantial partial result for arXiv:2601.10217, Question 3.8

11 August 2026

Status. Candidate substantial partial result, likely valid. The arbitrary-measure problem remains open, but the theorem below gives a complete criterion on an explicit infinite family of positive measures and produces nuclear examples of infinite total mass.

Source question

Ma, Lu, and Zu ask for a necessary and sufficient condition for a positive measure μ to induce a nuclear map $T_\mu : F_\alpha^p \rightarrow F_\alpha^q$ when $1 \leq p < q \leq \infty$ [1].

15

But $\int_{\mathbb{C}} f(z) \overline{g(z)} e^{-\alpha|z|^2} dA(z) \leq \int_{\mathbb{C}} |f(z)g(z)| e^{-\alpha|z|^2} dA(z)$. We obtain a contradiction.

Finally, we conclude this section with an open problem.

Question 3.8 For $1 \leq p < q \leq \infty$ and $\mu \geq 0$, what is the necessary and sufficient condition for $T_\mu \in \mathcal{N}(F_\alpha^p, F_\alpha^q)$?

4 Trace formula and density theorem

In this section, for an operator $T \in \mathcal{N}(F_\alpha^p)$, ($1 < p < \infty$), we will provide some formulas for calculating its trace. Finally, we will establish a density theorem.

Lemma 4.1 Let $1 < p < \infty$ and $T \in \mathcal{N}(F_\alpha^p)$, then we have:

The screenshot is Question 3.8 on page 15 of the source PDF.

Proof intuition

Widely separated normalized Fock kernels behave like the unit-vector basis of ℓ^t , simultaneously for all $1 \leq t \leq \infty$. For points $z_j = jR$, the analysis-synthesis Gram matrix has entries $e^{-\alpha R^2(i-j)^2/2}$. If R is large, its off-diagonal part has norm less than one on every ℓ^t , so the Fock analysis and synthesis maps have bounded one-sided inverses.

An atomic Toeplitz operator on these points is therefore, up to these bounded maps, exactly the diagonal operator $D_c : \ell^p \rightarrow \ell^q$. Tong's classical diagonal theorem identifies nuclearity of D_c with an ℓ^s condition, where $1/s = 1 - 1/p + 1/q$. The inverse factorization makes this condition necessary as well as sufficient. Since $s > 1$ away from the extreme endpoint, the criterion allows non-summable weights and hence measures of infinite mass.

Statement

Fix $\alpha > 0$ and $1 \leq p < q \leq \infty$, and define

$$\frac{1}{s} = 1 - \frac{1}{p} + \frac{1}{q}. \quad (1)$$

Thus $1 < s < \infty$ unless $(p, q) = (1, \infty)$, when $s = \infty$. Choose

$$R > \sqrt{\frac{2 \log 3}{\alpha}}, \quad z_j = jR \quad (j \geq 1), \quad (2)$$

and let $c_j \geq 0$. Set $\mu = \sum_{j \geq 1} c_j \delta_{z_j}$.

Theorem 1 (exact sparse-atomic criterion). *The measure μ satisfies the defining kernel-integrability condition of [1] whenever c belongs to the space below, and*

$$T_\mu \in \mathcal{N}(F_\alpha^p, F_\alpha^q) \iff \begin{cases} c \in \ell^s, & (p, q) \neq (1, \infty), \\ c \in c_0, & (p, q) = (1, \infty). \end{cases}$$

Moreover, with constants depending only on p, q, α, R ,

$$\nu(T_\mu) \asymp \begin{cases} \|c\|_{\ell^s}, & (p, q) \neq (1, \infty), \\ \|c\|_{\ell^\infty}, & (p, q) = (1, \infty), \quad c \in c_0, \end{cases}$$

where ν denotes the nuclear norm.

Analysis, synthesis, and the Gram inverse

The normalized kernel is

$$k_w(z) = e^{\alpha \bar{w}z - \alpha |w|^2/2}.$$

For $1 \leq t \leq \infty$, define

$$A_t f = (f(z_j) e^{-\alpha |z_j|^2/2})_{j \geq 1}, \quad S_t a = \sum_{j \geq 1} a_j k_{z_j}.$$

Lemma 2. $A_t : F_\alpha^t \rightarrow \ell^t$ and $S_t : \ell^t \rightarrow F_\alpha^t$ are bounded. Their composition $G_t = A_t S_t$ is invertible on ℓ^t . Consequently

$$R_t := S_t G_t^{-1} \quad \text{satisfies} \quad A_t R_t = I_{\ell^t},$$

and

$$L_t := G_t^{-1} A_t \quad \text{satisfies} \quad L_t S_t = I_{\ell^t}.$$

Proof. Choose $0 < \rho < R/2$. The standard submean estimate, applied after a Fock translation, gives for $t < \infty$

$$|f(z_j)|^t e^{-t\alpha |z_j|^2/2} \leq C_{t,\alpha,\rho} \int_{D(z_j,\rho)} |f(z)|^t e^{-t\alpha |z|^2/2} dA(z).$$

The disks are disjoint, so summation proves boundedness of A_t . For $t = \infty$ it follows directly from the Fock norm.

For synthesis, pointwise

$$|S_t a(z)| e^{-\alpha |z|^2/2} \leq \sum_j |a_j| e^{-\alpha |z - z_j|^2/2}.$$

Integration gives the $\ell^1 \rightarrow F_\alpha^1$ bound, while the uniformly bounded Gaussian sum gives the $\ell^\infty \rightarrow F_\alpha^\infty$ bound. Interpolation (or the same discrete convolution estimate) gives all intermediate t .

Because all z_j are real,

$$(G_t a)_i = \sum_{j \geq 1} e^{-\alpha R^2 (i-j)^2/2} a_j.$$

Put $u = \alpha R^2/2 > \log 3$. The off-diagonal row and column sums are at most

$$2 \sum_{n \geq 1} e^{-un^2} \leq \frac{2e^{-u}}{1 - e^{-u}} < 1.$$

Young's inequality (including its endpoint forms) therefore gives $\|G_t - I\|_{\ell^t \rightarrow \ell^t} < 1$ for every t . The Neumann series supplies G_t^{-1} , and the one-sided inverse identities follow. $\square$

Proof of the classification

Proof of Theorem 1. First, the relevant coefficient spaces are contained in ℓ^∞ . For each fixed $z \in \mathbb{C}$, completing the square shows

$$\sum_j c_j |K_\alpha(z, z_j)| e^{-\alpha|z_j|^2} \leq \|c\|_\infty e^{\alpha(\operatorname{Re} z)^2/4} \sum_j e^{-\alpha(z_j - \operatorname{Re} z/2)^2} < \infty.$$

Thus T_μ is defined on the kernel span.

Let $D_c : \ell^p \rightarrow \ell^q$ be the diagonal map $D_c(a_j) = (c_j a_j)$. Direct substitution into the definition of T_μ gives the exact factorization

$$T_\mu = \frac{\alpha}{\pi} S_q D_c A_p. \quad (3)$$

Tong's diagonal nuclearity theorem [2, 3] says

$$D_c \in \mathcal{N}(\ell^p, \ell^q) \iff \begin{cases} c \in \ell^s, & (p, q) \neq (1, \infty), \\ c \in c_0, & (p, q) = (1, \infty), \end{cases}$$

and its nuclear norm is the displayed sequence norm. Hence the stated condition implies nuclearity of T_μ by (3) and the operator ideal property.

Conversely, Lemma 2 and (3) give

$$D_c = \frac{\pi}{\alpha} L_q T_\mu R_p.$$

If T_μ is nuclear, so is D_c , and Tong's theorem gives the necessary sequence condition. Applying the nuclear ideal norm inequality to both factorizations yields the two-sided norm estimate. $\square$

Corollary 3 (positive infinite-mass nuclear symbols). *For every $1 \leq p < q \leq \infty$, there is a positive measure μ with $\mu(\mathbb{C}) = \infty$ such that $T_\mu \in \mathcal{N}(F_\alpha^p, F_\alpha^q)$.*

Proof. If $s < \infty$, choose a with $1/s < a \leq 1$ and set $c_j = j^{-a}$; then $c \in \ell^s \setminus \ell^1$. At $(p, q) = (1, \infty)$ take $c_j = 1/j$, which belongs to $c_0 \setminus \ell^1$. Apply Theorem 1. $\square$

Scope, verification, and novelty

The source proves that finite total mass is sufficient for all p, q and is necessary when $q \leq p$. Corollary 3 proves that this criterion cannot extend to $p < q$. The exponent in (1) also matches the source's necessary L^s Berezin-transform condition. The proof extends to any normalized-kernel sequence for which the corresponding analysis and synthesis maps are complemented; arbitrary non-atomic and densely atomic measures remain open because their local cell operators need not reduce to scalar diagonal blocks.

All normalization constants and inverse identities were checked directly; no computation is used. A bounded search on 11 August 2026 covered the run indexes, the exact wording of Question 3.8, the July 2026 accepted manuscript, and keyword searches for nuclear positive-measure Fock Toeplitz maps in the $p < q$ range. No later answer was found. Tong's theorem is classical prior background; novelty confidence for the Fock reduction and infinite-mass consequence is moderate-high.

Human-review recommendation. Check the endpoint statement of Tong's theorem and the factor α/π in (3). The Gram estimate is uniform in t and includes both endpoints.

References

- [1] T. Ma, Y. Lu, and C. Zu, *Nuclear Toeplitz Operators between Fock Spaces*, arXiv:2601.10217; accepted manuscript, Canadian Math. Bull. (2026), doi:10.4153/S0008439526102227.
- [2] A. Tong, *Diagonal nuclear operators on ℓ_p spaces*, Trans. Amer. Math. Soc. 143 (1969), 235–247.
- [3] D. D. Haroske, H.-G. Leopold, and L. Skrzypczak, *Nuclear embeddings in general vector-valued sequence spaces with an application to Sobolev embeddings of function spaces on quasi-bounded domains*, arXiv:2009.00474, Proposition 2.2.